



PoliKick

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1 Introduction

The project aims to emulate the famous Roland 808 kick, based on a schematic inspired by the one proposed by Erika Synths. The kick sound is obtained by accurately reproducing its different parts, whose characteristics we want to be adjustable via knobs to achieve the desired behavior.

2 Requirements

The kickdrum is the foundation of a beat. It's the pulse of a track, creating rhythm and energy. A kick can be short and punchy or long and booming, it often has a deep bass presence or a clear pitch. To reproduce these characteristic features of a kick, it is necessary to understand its fundamental parts

2.1 The structure of the Kick sound

- **Transient (Attack):** The part that generates the initial click, consisting of higher frequencies.
- **Body:** Defines the tonal character of the kick and corresponds to the steady-state frequency.
- **Decay:** Controls the duration of the sound before it fades away.

2.1.1 Attack

The Kick frequency should start at a high initial value and quickly decrease until it stabilizes at a "steady-state" value in order to achieve a punchy sound.

We want to integrate parameters within the circuit that allow us to adjust:

- Initial frequency
- Depth of the pitch transient
- Frequency Transient duration

2.1.2 Body

Once the transient phase has ended, the frequency will settle at a value that remains constant over time.

The tonal character of the 808 kick is given by the frequency reached after the transient and by the amplitude and the distortion achieved by the signal.

We want to include parameters within the circuit that allow us to adjust:

- Steady-state frequency

-
- Distortion
 - Tone

2.1.3 Decay

The decay models the impulsive nature of the kick, determining how quickly the sound fades away.

We want to include parameters within the circuit that allow us to adjust:

- Sound duration

2.2 User Parameters

In the following graph, we observe two different kick drum sounds, each with a distinct decay profile.

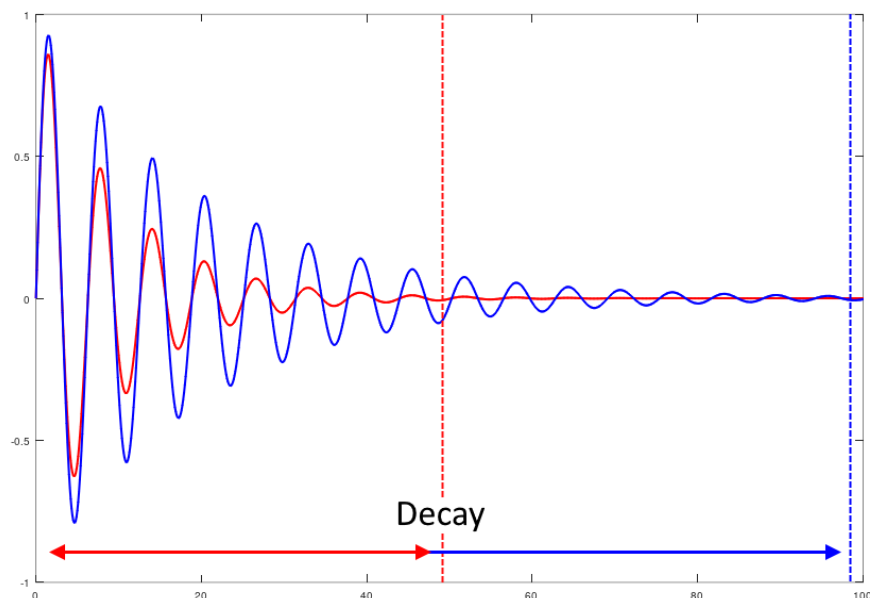


Figure 2.1

To shape the **decay**, we can adjust the *decay* parameter accordingly. Increasing its value makes the sound fade away slower.

In the next image, a graph shows the frequency response over time.

We can clearly distinguish between the attack and the body of the sound. The body section, in particular, exhibits a constant frequency, known as the *steady-state frequency*.

In the following figure, we can identify additional frequency-related characteristics.

These aspects can be shaped using specific parameters:

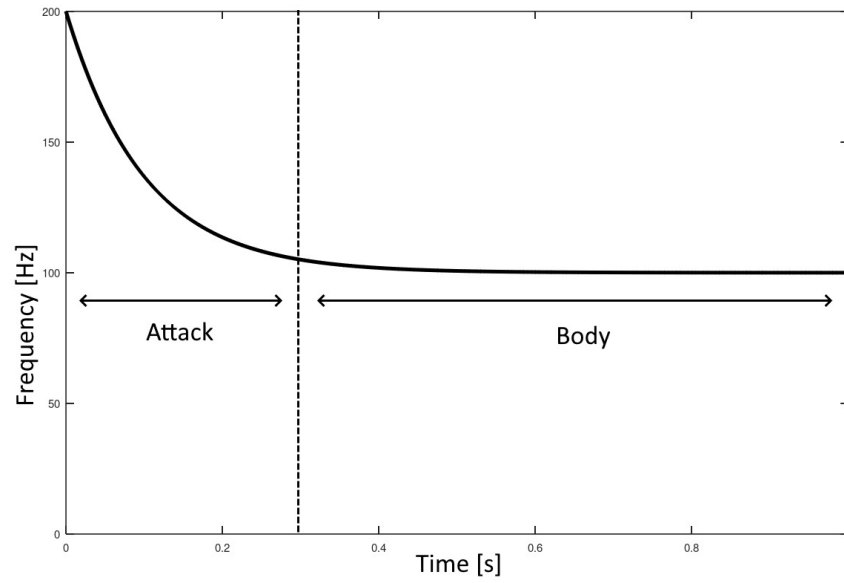


Figure 2.2: Frequency over time

- Transient Duration
- Steady-State Frequency
- Initial Frequency

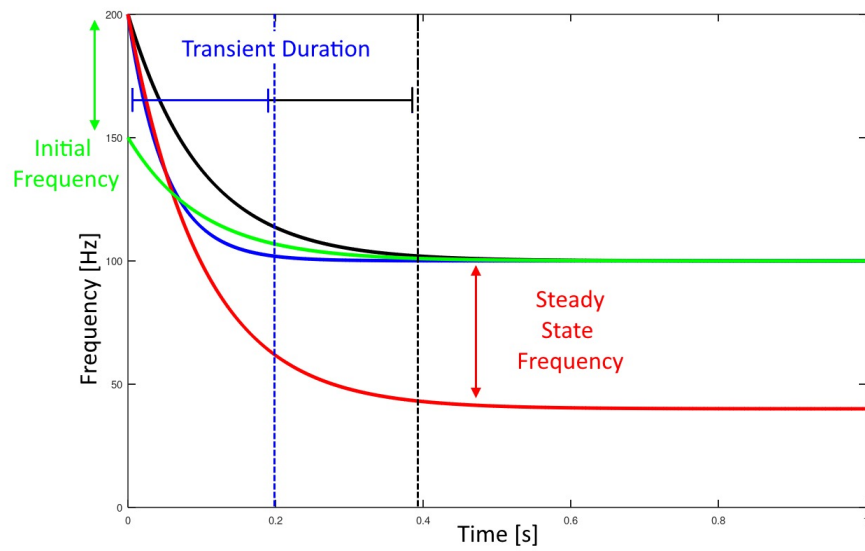


Figure 2.3: Frequency over time

To increase the **Initial Frequency** we can:

- increase the value of TuneDepth
- increase the value of Pitch

To increase the **Steady-State Frequency** we can:

- increase the value of Pitch
- increase the value of PitchCV Amount

To increase the **Transient Duration** we can:

- increase the value of TuneDecay

2.2.1 AccentCV

When the AccentCV voltage drops below 2 V, other aspects of the circuit behavior—such as transient and steady-state frequencies—are lightly affected.

3 Schematic

4 Stages

4.1 Gate to Trigger

Let's start from the very first block of the circuit. The purpose of the *gate to trigger* is to transform the signal sent from the sequencer, which is a succession of 'gates', into a sequence of short impulses which are used at a later stage to drive the *T-oscillator* properly.

4.1.1 Analysis

The input voltage v_i is the sequencer signal. It is a sequence of rectangular windows (gates).

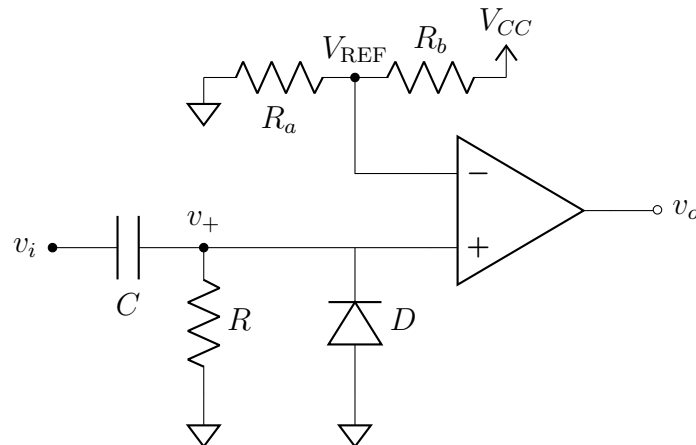


Figure 4.1: Gate to Trigger

The signal v_i goes first of all through an RC high-pass filter, hence v_+ takes on the characteristic exponential shape upon every rising or falling edge of v_i . The presence

of D is needed because some op-amps undergo an effect called "phase inversion" if the voltage at either terminal becomes too negative; the diode blocks v_+ at a fixed value during the falling edge of v_i (figure ??).

The op-amp is in a non inverting comparator configuration (without hysteresis):

$$v_o = \begin{cases} +V_{CC}, & v_A > V_{REF} \\ -V_{CC}, & v_A < V_{REF} \end{cases} \quad V_{REF} = kV_{CC} \quad k \triangleq \frac{R_a}{R_a + R_b}$$

The cut-off frequency of the high-pass filter is $f_0 = \frac{1}{2\pi\tau}$, where $\tau = RC$. During the rising edge of v_i , we get $v_A > V_{REF}$ for a small time Δt . During this interval the output of the comparator is V_{CC} . Let's examine what happens during the transient:

$$\begin{aligned} v_A(t) > V_{REF} &\implies V_I e^{-t/\tau}(t) > V_{REF} = kV_{CC} \geq 0, \quad t \geq 0 \\ &\implies 0 < t < \tau \log \frac{V_I}{kV_{CC}} \triangleq \Delta t \end{aligned} \quad (4.1)$$

In order to have a working circuit we need to ensure that the slew rate of the op-amp be sufficiently high:

$$S_R \gg \frac{2V_{CC}}{\Delta t}$$

Futhermore, the presence of the op-amp guarantees a small output impedance $R_{out} \approx 0$.

Example 1. Let's choose $\Delta t = \frac{T}{10}$ where T is the gate duration. We need to make sure that τ is sufficiently small compared to T , e.g. $\tau = \Delta t = \frac{T}{10}$:

$$\begin{aligned} \tau \log \frac{V_I}{kV_{CC}} = \Delta t &\implies V_I = e \cdot kV_{CC} \implies k = \frac{R_a}{R_a + R_b} = \frac{V_I}{eV_{CC}} \\ \tau = RC &\implies C = \tau/R = \frac{T}{10R} \end{aligned}$$

4.1.2 Summary

- $v_i(t) = V_I, \quad 0 \leq t \leq T$
- $V_{REF} = \frac{R_a}{R_a + R_b} V_{CC}$
- $\Delta t = \tau \log \frac{V_I}{V_{REF}}$
- $S_R \gg \frac{2V_{CC}}{\Delta t}$

4.2 Accent Control Voltage

The *Accent CV* subcircuit is used to control the amplitude of the impulse generated by the *gate to trigger* block by means of an external control voltage v_A (figure 4.2).

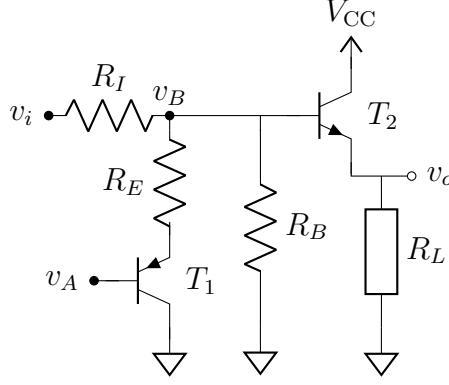


Figure 4.2: Accent CV

4.2.1 Analysis

First of all we notice that T_1 cannot saturate because $V_{CB,1} = -v_A \leq 0$ V, hence the collector-base junction does not allow current to flow.

Then, since the input voltage v_i is the output of the *gate to trigger*, we can consider two different cases:

1. Impulse (comparator output is high): $v_i = +V_{AL}$
2. Rest (comparator output is low): $v_i = -V_{AL}$

For the first case, we suppose that T_1 e T_2 are in the active linear region. We need to make sure that this hypothesis is compatible with the results. T_2 is in a voltage follower configuration: the voltage v_B is copied, with a small offset, to the output v_o . Similarly, we can observe that the voltage on the emitter of T_1 is the same as v_A with an offset; as such, T_1 behaves like a voltage source $v_A + V_{EB,1,on}$.

Supposing that the base current of T_2 be negligible, we obtain the following expression for v_B :

$$v_B = \frac{R_I \parallel R_B}{R_E + R_I \parallel R_B} (v_A + V_{EB,1,on}) + \frac{R_E \parallel R_B}{R_I + R_E \parallel R_B} V_{AL} \quad (4.2)$$

The output of this stage is $v_o = v_B - V_{BE,2,on}$.

In the second case $v_i = -V_{AL}$, so $v_B < 0$; both transistors are off and the output is $v_o = 0$ V.

4.3 T-Oscillator

The *T-Oscillator* (figure 4.3) features one op-amp, along with two resistors and capacitors; together they behave like a quite unique sinusoidal oscillator which needs an incoming impulse to start. We use an impulse (instead of a gate) to control v_+ in order to center the oscillations around 0 V.

This circuit employs a particular bridged-T network, which behaves as a band-pass filter when placed along the negative feedback loop of the op-amp. However, once

the oscillations have started, the output voltage amplitude will gradually decay in an exponential fashion due to the bridge resistance. This behavior is often undesirable for an oscillator, but it is perfect for a kick drum.

It is possible to derive an expression for the transfer function of the circuit, although its parameters depend on the other input voltages applied externally. The expression takes the following form:

$$H(s) = \frac{a_2 s^2 + a_1 s + 1}{b_2 s^2 + b_1 s + 1}$$

From this we can derive some important parameters: the central frequency and the quality factor of the band-pass filter, f and Q .

$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad (4.3)$$

$$Q = \frac{\sqrt{\frac{R_1}{R_2}}}{\sqrt{\frac{C_1}{C_2} + \frac{C_2}{C_1}}} \quad (4.4)$$

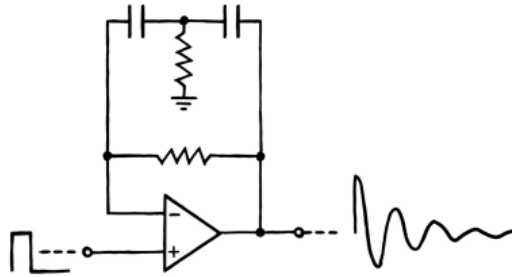


Figure 4.3: T-Oscillator circuit

To properly control the frequency and the decay of the oscillation, we need to choose the right combination of values for the capacitors and the resistors. This turns out to be somewhat complicated since changing any value alters both the frequency and the decay rate.

In the original 808 kick, Roland used a bridge resistance of $1\text{ M}\Omega$, a resistance to ground of $51\text{ k}\Omega$, leading to 50 Hz oscillations and a very quick decay. We use these values as a starting point.

4.4 Amplitude EG

The model used is the one shown in the figure ???. The basic idea is to derive an analytical expression for the decay, which will be adjusted via a potentiometer.

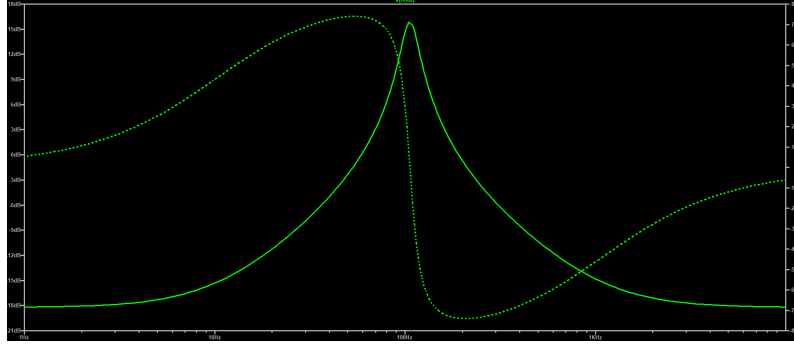


Figure 4.4: T-oscillator

4.4.1 Decay and frequency dependency

The main issue with the T-bridge is that frequency and decay are not independent values. The central resistance R_S influences both values. Additionally, it must be considered that, apart from R_s , the values are also affected by the overall resistance seen between node F and ground. Specifically, R_S is in parallel with R_4 (whose value is approximately comparable to 1 M Ω) and with R_3 . In the MATLAB script, for simplicity, Rf was considered as $R_S \parallel R_4$. To address this dependency issue, an inverting amplifier was introduced, whose absolute gain was named G_i). By using this parameter, the crosstalk between the two can be minimized (the resistance at node F does not depend on G_i).

4.4.2 Analytic expression

As can be seen from the obtained function (and as one might expect), the signal is composed of two parts: an oscillating component (the sound) multiplied by a negative exponential (the decay). The time constant (considering $C_1 = C_2$) of the exponential has a value of:

$$\tau = (2C_1 R_3 R_B) / (2R_3 - G_i R_B) \quad (4.5)$$

For the one of interest, we can consider the kick to last approximately 3 to 4 times τ .

4.4.3 Code

```

1  % This code evaluates the equation of the T-bridge inserted in the
2  % rejection ring.
3  % Rf is the parallel value of Rs and R4
4  % that in this evaluation is considered as a constant value
5  close all
6  clear
7  format shortE
8  syms Vo Vi Vf G_i Rf R3 Rb C1 C2 S t k;
9  % NOTE: G_i is a positive value!!
10
11 eq1 = (Vi-Vf)*S*C2 == (Vo-Vi)/Rb;
12 Vf = solve(eq1, Vf); % solves eq1 for Vf
13 eq2 = Vf./Rf == (Vi-Vf).*S*C2 + (Vo-Vf)*S*C1 - (Vo*G_i + Vf)./R3;
```

```

14 sol_Vo = solve(eq2, Vo)/Vi; % solves eq2 for Vo
15 Vo_final_s = subs(sol_Vo, C2, C1); % substitutes C2 with C1 (same value)
16 pretty(collect(Vo_final_s, S)); % prints the transfer function collecting
    ↪ S
17 pretty(ilaplace(Vo_final_s, S, t)); % prints the time response
18
19 tau = (2*C1*R3*Rb) / (2*R3 - G_i*Rb); % symbolic expression of Tau
20 tau_n = subs(tau, [Rf, R3, Rb, C1], [53*10^3, 470*10^3, 1*10^6,
    ↪ 15*10^(-9)]); % numeric expression of tau
21 decay = vpa(5*tau_n); % We assume the signal fades after 5 tau
22
23 % Substitute G_i in function of k, position of the potentiometer
24 Gi = (500*47*10^6*k / ((500*k + 47)*10^3)) / (47*10^3);
25 decay_lin = subs(decay, G_i, Gi); % makes the decay more linear
26
27 figure;
28 fplot(decay, [0, 0.92])
29 xlabel('Gi');
30 ylabel('decay');
31 title('Decay based on Gi');
32
33 figure;
34 fplot(decay_lin, [0.2, 1])
35 xlabel('k');
36 ylabel('decay');
37 title('Decay based on potentiometer position');

```

4.5 Pitch EG

We aim to reproduce the transient of the kick: the frequency must start from a high value and then decrease, settling at a steady-state frequency.

To achieve this effect, we place a parallel branch to the T-Oscillator resistance, where we implement a BJT transistor (Q5 in the image) whose equivalent resistance between the collector and the emitter will be varied.

A capacitor gives the transient behavior, the voltage across it is reported onto the emitter of the transistor Q4, which controls the operating point of the transistor Q5.

At the start Q5 is fully saturated, acting as a short circuit, after the transient it behaves as an open circuit.

4.5.1 Pitch

The frequency is determined by the resistance within the T-Oscillator.

Through the Pitch control, it is possible to modify both the transient and steady-state frequency.

The effect it causes is hyperbolic in nature, therefore $f_{steady-state} \propto \frac{1}{R_{Pitch}}$ (figure 4.5).

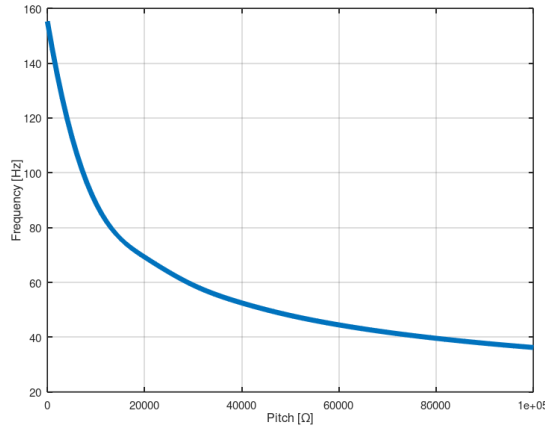


Figure 4.5: Frequency as a function of Pitch

4.5.2 TuneDepth

In the parallel branch, we implement not only the BJT transistor but also a fixed resistor (TuneDepth). When the equivalent resistance between the emitter and the collector of Q5 is low the frequency is controlled by the TuneDepth.

The Q5 transistor operating point is controlled by a capacitor, which charges quickly during the initial pulse and then discharges following an exponential behavior (RC discharge).

- When the capacitor is charged, the transistor is fully saturated, and its equivalent resistance is low. The frequency is mainly controlled by the TuneDepth resistance.
- When the capacitor is discharged, the branch behaves like an open circuit.

The effect it causes is again hyperbolic in nature, so $f(t) \propto \frac{1}{R_{TuneDepth}}$

Note that if PitchCV is sufficiently high, then TuneDepth also affects the steady-state frequency.

4.5.3 Pitch CV

As mentioned earlier, the voltage across the capacitor decreases exponentially (RC discharge). The voltage value will continue to drop until it stabilizes approximately at the voltage set by the voltage divider on PitchCV (plus 0.6 V due to the voltage drop across the diode of Transistor Q3).

If PitchCV is high, the capacitor will not fully discharge, and transistor Q5 will remain partially saturated (TuneDepth will continue to affect the frequency even at steady-state, as Q5 will not reach cutoff and will not act as an open circuit).

It is worth noting the oscillatory behavior of the voltage at node 3. This behavior changes the operating region of the transistor during oscillation and slightly distorts the sinusoid, adding harmonics.

With the potentiometer at 50%, PitchCV significantly increases the steady-state frequency in the range between 0.5 V and 3 V. As a first approximation, in this range the steady-state frequency as a function of PitchCV can be considered linear.

If the Pitch resistance has a low value, then the effect of PitchCV becomes less significant (figure 4.6).

Finding an analytical function for the settling frequency is not trivial, so we proceeded through simulation, and we obtained the following graph.

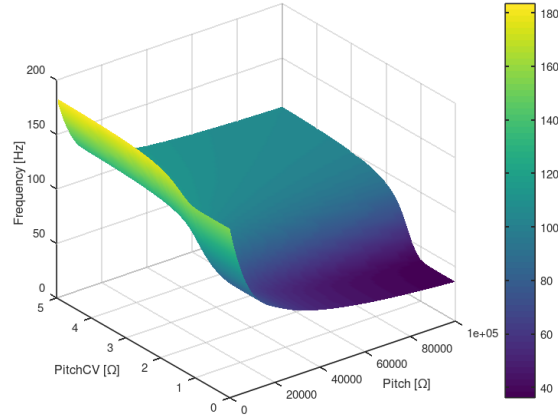


Figure 4.6: Frequency as a function of Pitch and PitchCV

4.5.4 TuneDecay

TuneDecay is the resistance that determines the discharge speed of the capacitor. Increasing it will cause a longer transient duration.

The formula for the time constant is approximately given by:

$$\tau = (R_{\text{tune decay}} + 10 \text{ M}\Omega) \cdot 220 \text{ nF}$$

4.5.5 Effects on Transient and steady state frequency

Below is the effect of various control parameters on the steady-state frequency and transient frequency:

Affects the frequency in terms of		
	Steady-state	Transient
PitchCV	Yes	No
TuneDepth	If $PitchCV > 2 \text{ V}$ with potentiometer = 50%	Yes
Pitch	Yes	Yes

Table 1: Effect of PitchCV, TuneDepth, and Pitch on steady-state and transient frequency

4.5.6 Implementation and Simulation

For the simulation, only the steady-state frequency was considered. The period duration was measured after 200 ms, once the transient was extinguished.

Multiple observations were conducted using different values of PitchCV, TuneDepth, and Pitch. From the graphs obtained using MatLab, it can be observed that PitchCV and Pitch cause independent variations in the steady-state frequency. However, it is noteworthy that TuneDepth and PitchCV are correlated in terms of modifying the steady-state frequency.

4.5.7 Frequency dependence on AccentCV

Indirectly, AccentCV also causes variations in frequency.

Simulations have shown a correlation between PitchCV and AccentCV.

Depending on the value of AccentCV, the capacitor will charge differently. If AccentCV has a low voltage value and PitchCV is high, then once the capacitor is charged, it will not be able to discharge, and the frequency will remain constant for the entire duration of the kick.

If AccentCV is not sufficiently high, the capacitor will not charge enough to drive transistor Q5 into saturation and make the TuneDepth resistance visible.

To avoid frequency dependence on AccentCV, it is necessary to work with a voltage higher than approximately 2.2 V.

4.6 Low Pass Filter

The second last stage is a simple low-pass filter. Its transfer function is given by the following expression.

$$H(s) = \frac{1}{sCxP + 1} \quad (4.6)$$

What is obtained, basically, is a *smoothing* or an *averaging* of the signal in time, since the voltage on the capacitor doesn't follow the input voltage exactly but it charges and discharges, with a time constant which is the product of the resistance and of the capacity of the capacitor.

$$\tau = xPC \quad (4.7)$$

In frequency domain, it means selecting a certain range of frequencies, in this case the lower ones (between 0 and 300 Hz). According to circuit theory, the capacitor acts like an open circuit at low frequency, while it becomes a short to *GND* at high frequency, annihilating the signal. This is compliant with the project requirements, since the idea is to produce an 808 Kick (hence a Bass). Notably, from the τ equation, the time constant and, consequently, the cut-off frequency f_c , is tunable to taste with the P potentiometer: rotating the knob to reduce the value of P also reduces τ , which increases f_c . This means that the frequency interval is wider and "catches" higher frequencies.

4.7 Distortion

4.7.1 Basic concepts

The output stage is a distortion stage. Distortion, generally speaking, means loss of information on the signal. In this case, it consists in cutting or limiting in magnitude the signal or, more precisely, to (*clamp* or *clip*, in musicmaking terms).

In the auditory field, the effect of distortion is to increase the power of the sound (the volume, in practice) while making it harsh-sounding. This effect, electronically speaking, is obtained with two elements: an amplifier and some diodes.

The behavior of the diodes, because of their physical composition, can be idealized as a short or an open circuit, if current flows through them or not; actually, when current flows, there's a certain voltage drop on the diode (usually between 0.5 V and 1.2 V, depending on the diode), which is called *threshold voltage* (or just *threshold*). What happens is that the diode, in this configuration, stops the signal amplitude at its *threshold*. The purpose of the amplifier is to increase the amplitude of the signal in order to exceed the *threshold* and, this way, to get *clamped*.

The circuit in exam was design to achieve the so-called *soft clip* effect (opposing to the *hard clip* effect, which is not discussed here), with the ability to switch it on and off. The *clamp* block is made by two antiparallel diodes, in order to let the current flow through that circuit branch in both directions, independently from the sign of the voltage. The amplification block is a non-inverting amplifier.

The potentiometer controls the gain of the amplifier, which also emphasizes (or attenuates) the distortion effect. Specifically, with the knob turned all the way to the left (minimum value), the amplifier assumes a *voltage follower* configuration, so $v_o = v_i$, which means it doesn't amplify the incoming signal.

The resistance R_C has been chosen to set the gain.

The behavior of the circuit will be analyzed in the following paragraphs.

4.7.2 Soft Clip

- **Diodes OFF**

For output voltages lower than the threshold voltage, v_D , plus the voltage on the resistance, *clipping* doesn't happen and the signal is untouched.

$$v_o = \left(1 + \frac{xP}{R_C}\right) v_i \quad (4.8)$$

where x is a coefficient between 0 and 1 which represents the cursor position of the knob, possibly in a nonlinear (e.g. logarithmic) fashion.

- **Diodes ON**

When the output voltage is greater than $v_D + i_C R_s$, where $i_C = \frac{v_i}{R_C}$, the diodes start conducting and limiting the voltage. If $xP \gg R_s$, it's reasonable to consider that i_c goes through R_s and the diodes only. As a consequence

$$v_o = i_c(R_C + R_s) + v_D = \left(1 + \frac{R_s}{R_C}\right) v_i + v_D \quad (4.9)$$

While the *soft clip* is inactive, the circuit behaves accordingly to the **Diodes OFF** case.